Mobile Application Development
(CS-375)
Assignment : 02

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Full Stack Cafe Ordering Mobile Application

Introduction:

This project is a full-stack cafe ordering mobile application that allows customers to browse menu items, add items to cart, and place orders. The application also includes a manager dashboard to receive orders.

The system is divided into two parts:

- Frontend (Mobile Application)
- Backend (Server & API)

Technologies Used:

Frontend:

- React Native
- Expo
- Expo Router
- Axios (for API requests)

Backend:

- Node.js
- Express.js

System Structure:

The application follows a client-server architecture:

- Mobile app sends HTTP requests to backend.
- Backend processes data.
- Backend returns JSON response.
- Orders are sent to manager dashboard.

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Flow:

User → Mobile App → API → Server → Response → App

Features Implemented:

Customer Side:

✔ View cafe menu

✔ Modern gradient UI

✔ Add to cart

✔ Place order

Manager Side:

✔ Receive orders

✔ Dashboard screen

6. API Endpoints

GET /menu

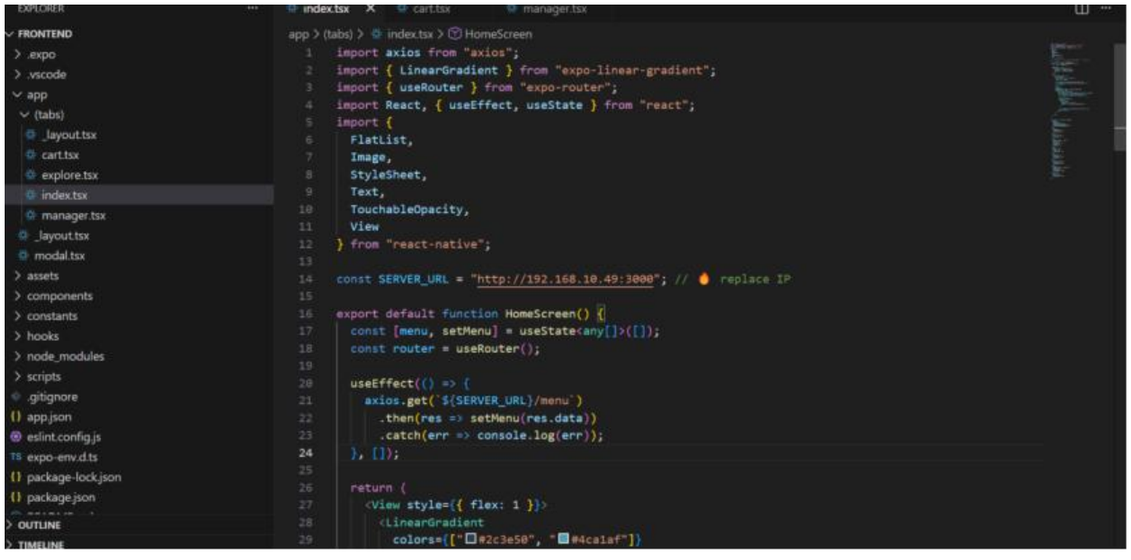
Returns list of café menu items.

POST /order

Receives order data including:

- Customer name
- Ordered items
- Total amount

Frontend Screenshots:



app > (tabs) > index.tsx > HomeScreen

```
16  export default function HomeScreen() {
36          <FlatList
37            data={menu}
38            keyExtractor={(item) => item.id.toString()}
39            renderItem={({ item }) => (
40              <View style={styles.card}>
41                <Image source={{ uri: item.image }} style={styles.image} />
42                <Text style={styles.name}>{item.name}</Text>
43                <Text style={styles.price}>Rs. {item.price}</Text>
44
45                <TouchableOpacity
46                  style={styles.button}
47                  onPress={() =>
48                    router.push({
49                      pathname: "/(tabs)/cart",
50                      params: { item: JSON.stringify(item) },
51                    })
52                  }
53                >
54                  <Text style={styles.buttonText}>Add to Cart</Text>
55                </TouchableOpacity>
56              </View>
57            )}
58          />
59        </View>
60      );
61  }
62
63  const styles = StyleSheet.create({
```

```
63  const styles = StyleSheet.create({
86    image: {
87      width: "100%",
88      height: 160,
89      borderRadius: 12,
90    },
91    name: {
92      fontSize: 18,
93      fontWeight: "bold",
94      marginTop: 10,
95    },
96    price: {
97      color: "#27ae60",
98      fontWeight: "bold",
99      marginVertical: 8,
100   },
101   button: {
102     backgroundColor: "#e67e22",
103     padding: 12,
104     borderRadius: 10,
105     alignItems: "center",
106   },
107   buttonText: {
108     color: "#fff",
109     fontWeight: "bold",
110   },
111 });
```

Cart.tsx:

app > (tabs) > cart.tsx > CartScreen

```
import axios from "axios";
import { useLocalSearchParams, useRouter } from "expo-router";
import { Alert, StyleSheet, Text, TouchableOpacity, View } from "react-native";

const SERVER_URL = "http://192.168.10.49:3000"; // same IP

export default function CartScreen() {
  const { item } = useLocalSearchParams();
  const router = useRouter();
  const parsedItem = item ? JSON.parse(item as string) : null;

  const placeOrder = () => {
    axios.post(`${SERVER_URL}/order`, {
      customerName: "Alina",
      items: [parsedItem.name],
      totalAmount: parsedItem.price,
    })
      .then(() => {
        Alert.alert("Order Placed!");
        router.push("/(tabs)/manager");
      })
      .catch(err => console.log(err));
  };

  if (!parsedItem) return null;

  return (
    <View style={styles.container}>
      <Text style={styles.title}>🛒 Cart</Text>
```

```
export default function CartScreen() {
      <Text style={styles.price}>Rs. {parsedItem.price}</Text>

      <TouchableOpacity style={styles.button} onPress={placeOrder}>
        <Text style={{ color: "#fff" }}>Place Order</Text>
      </TouchableOpacity>
    </View>
  );
}

const styles = StyleSheet.create({
  container: { flex: 1, justifyContent: "center", alignItems: "center" },
  title: { fontSize: 22, fontWeight: "bold", marginBottom: 20 },
  item: { fontSize: 18 },
  price: { fontSize: 16, marginVertical: 10 },
  button: {
    backgroundColor: "#27ae60",
    padding: 15,
    borderRadius: 10,
    marginTop: 20,
  },
});
```

Manager.tsx:

index.tsx cart.tsx manager.tsx ×

app > (tabs) > manager.tsx > ...

```
1   import { StyleSheet, Text, View } from "react-native";
2
3   export default function ManagerScreen() {
4     return (
5       <View style={styles.container}>
6         <Text style={styles.title}>👨‍💼 Manager Dashboard</Text>
7         <Text>Order received successfully!</Text>
8       </View>
9     );
10  }
11
12  const styles = StyleSheet.create({
13    container: { flex: 1, justifyContent: "center", alignItems: "center" },
14    title: { fontSize: 22, fontWeight: "bold", marginBottom: 15 },
15  });
```

Cmd:

```
C:\Users\City Laptop>cd Desktop

C:\Users\City Laptop\Desktop>cd assignment 2

C:\Users\City Laptop\Desktop\assignment 2>cd frontend

C:\Users\City Laptop\Desktop\assignment 2\frontend>npm start -- --clear

> frontend@1.0.0 start
> expo start --clear

Starting project at C:\Users\City Laptop\Desktop\assignment 2\frontend
React Compiler enabled
Starting Metro Bundler
warning: Bundler cache is empty, rebuilding (this may take a minute)

› Metro waiting on exp://192.168.10.49:8081
› Scan the QR code above with Expo Go (Android) or the Camera app (iOS)

› Web is waiting on http://localhost:8081
```

Backend Screenshots:

```js
const express = require('express');
const app = express();
const PORT = 3000;

app.use(express.json());

// ----------------------
// HARDCODED MENU DATA
// ----------------------
const menu = [
  {
    id: 1,
    name: "Margherita Pizza",
    price: 1200,
    category: "Daily Specials",
    image: "https://images.unsplash.com/photo-1601924582975-7e6f4e1f2c9c"
  },
  {
    id: 2,
    name: "Zinger Burger",
    price: 550,
    category: "Fast Food",
    image: "https://images.unsplash.com/photo-1550547660-d9450f859349"
  },
  {
    id: 3,
    name: "Pasta Alfredo",
    price: 900,
    category: "Italian",
```

```js
const menu = [
    id: 4,
    name: "Chocolate Shake",
    price: 350,
    category: "Beverages",
    image: "https://images.unsplash.com/photo-1572490122747-3968b75cc699"
  }
];

// ----------------------
// GET ENDPOINT (Catalog)
// ----------------------
app.get('/menu', (req, res) => {
  res.json(menu);
});

// ----------------------
// POST ENDPOINT (Order Logger)
// ----------------------
app.post('/order', (req, res) => {
  const order = req.body;

  console.log("===== NEW ORDER RECEIVED =====");
  console.log("Customer Name:", order.customerName);
  console.log("Items:", order.items);
  console.log("Total:", order.totalAmount);
  console.log("==============================");

  res.json({
```

Cmd:

```
C:\Users\City Laptop>cd Desktop

C:\Users\City Laptop\Desktop>cd assignment 2

C:\Users\City Laptop\Desktop\assignment 2>cd server

C:\Users\City Laptop\Desktop\assignment 2\server>node app.js
Server running on http://localhost:3000
===== NEW ORDER RECEIVED =====
Customer Name: Alina
Items: [ 'Zinger Burger' ]
Total: 550
==============================
===== NEW ORDER RECEIVED =====
Customer Name: Alina
Items: [ 'Pasta Alfredo' ]
Total: 900
==============================
```

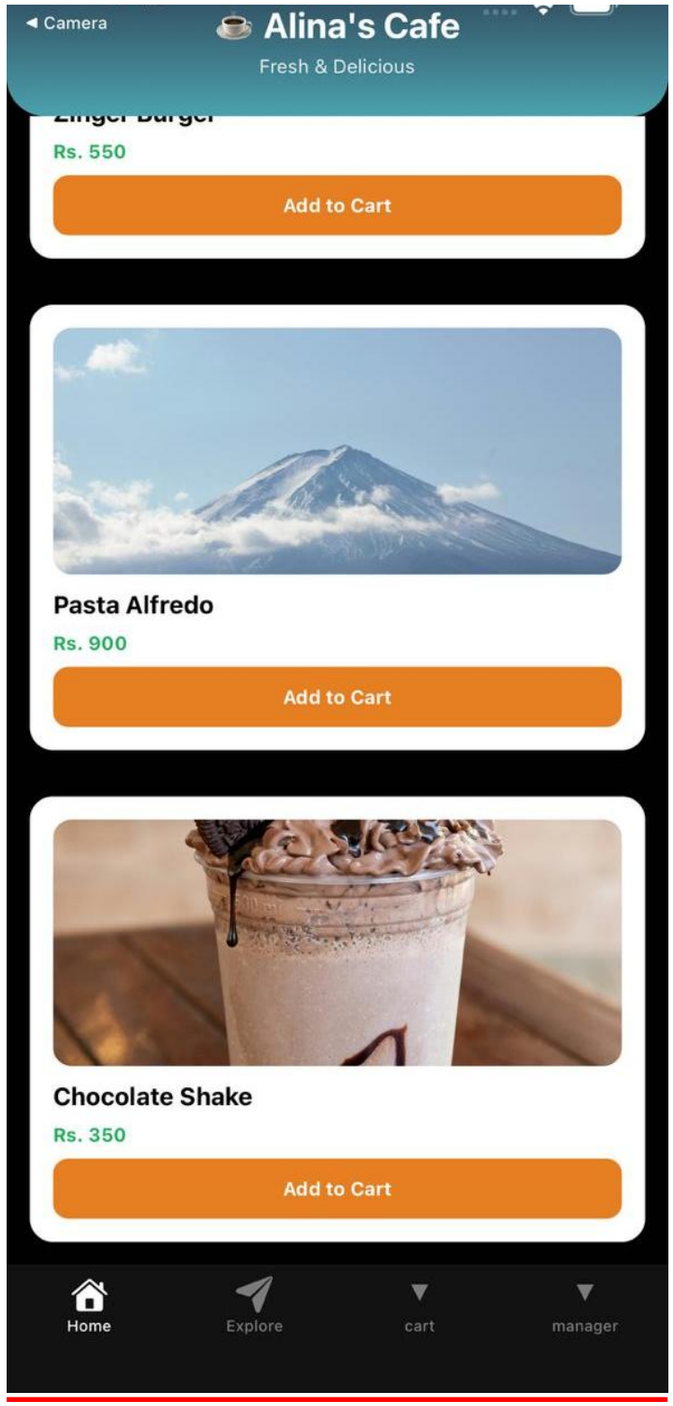
Menu:

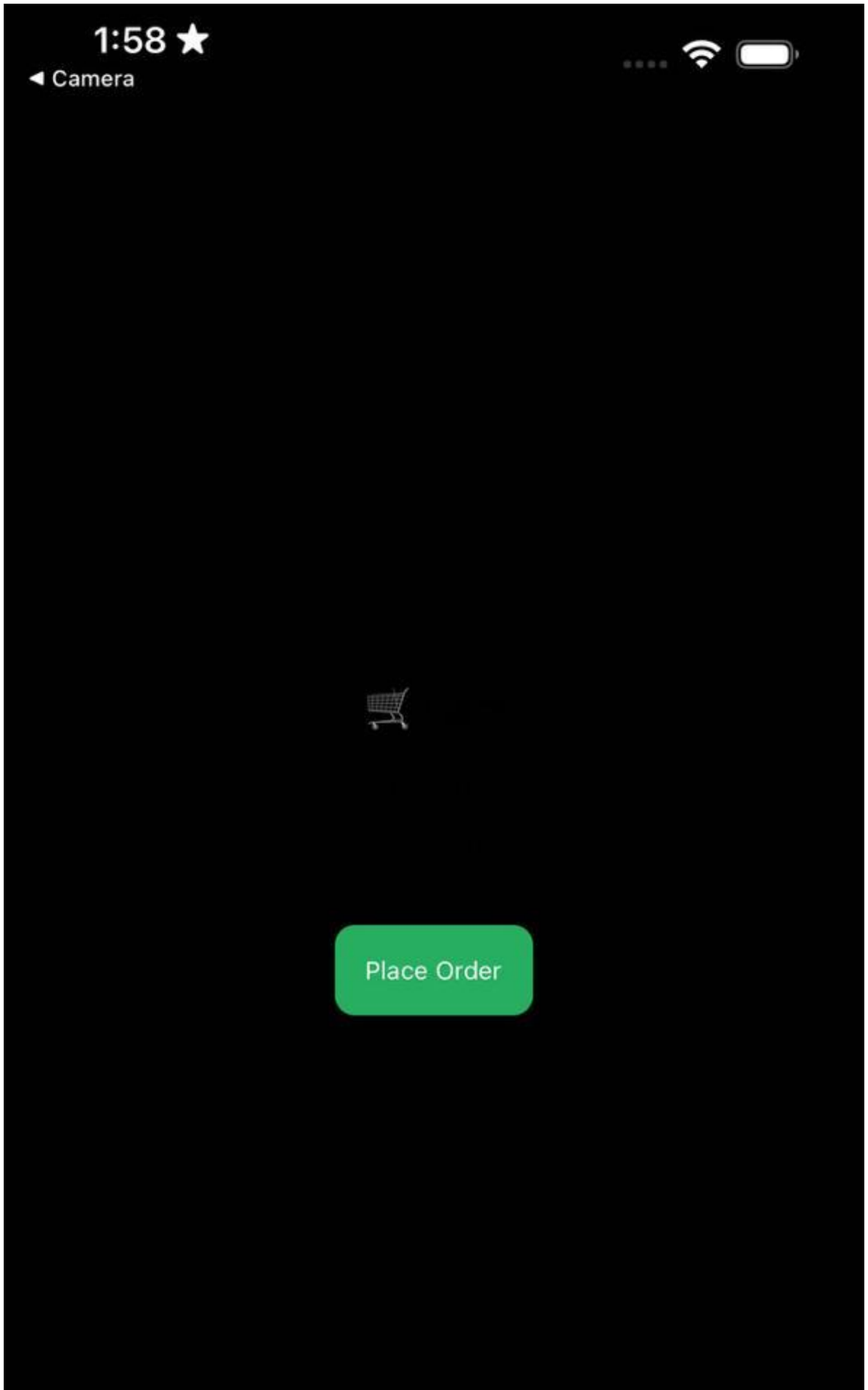
localhost:3000/menu

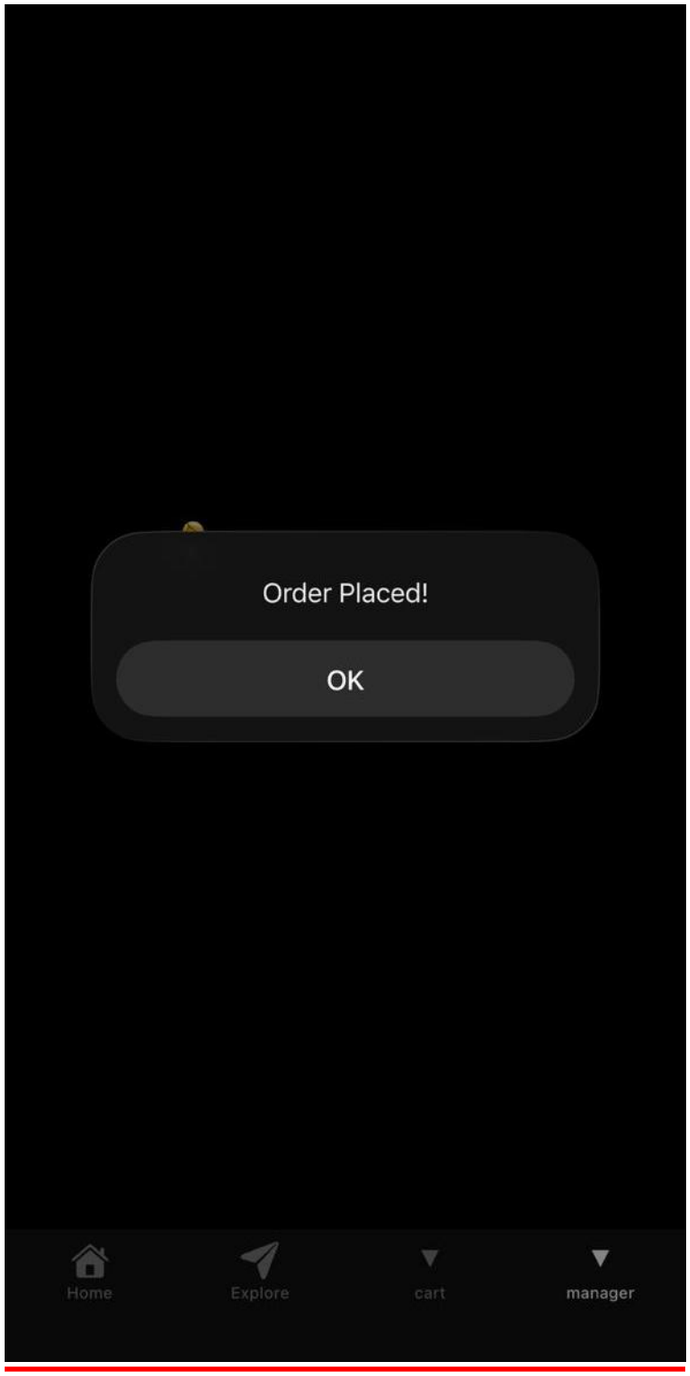
Pretty print

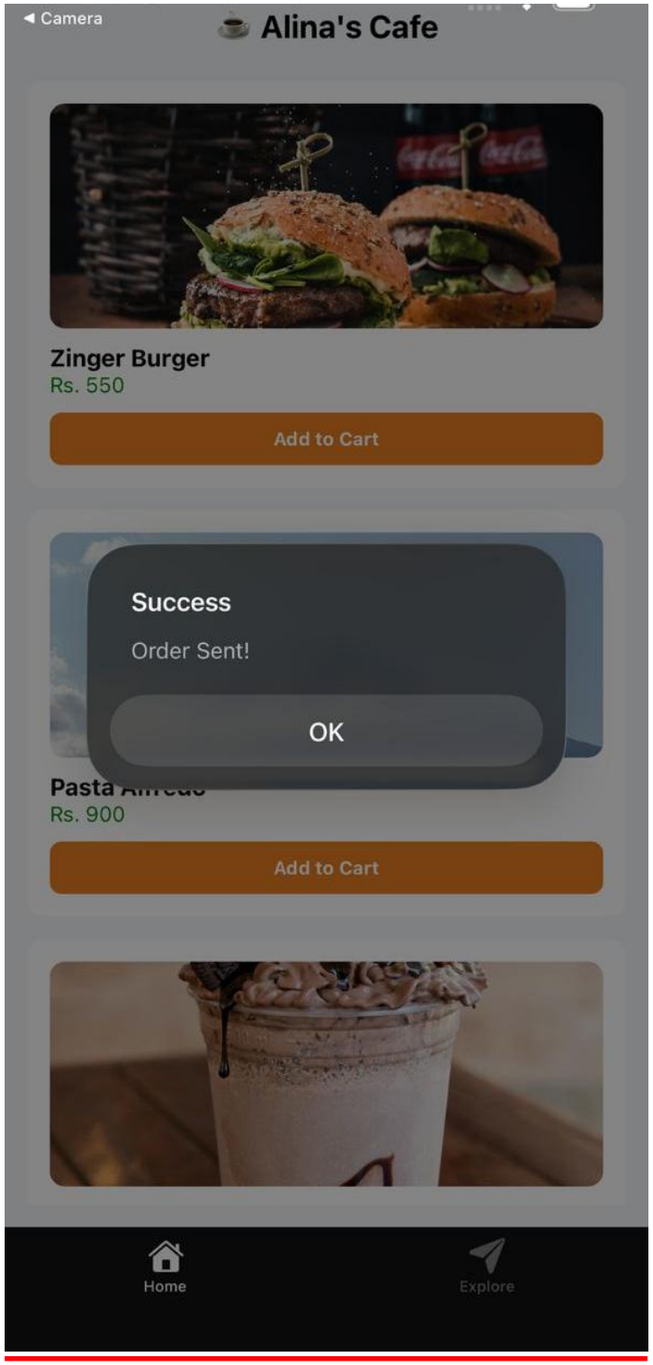
[{"id":1,"name":"Margherita Pizza","price":1200,"category":"Daily Specials","image":"https://images.unsplash.com/photo-1601924582975-7e6f4e1f2c9c"},{"id":2,"name":"Zinger Burger","price":550,"category":"Fast Food","image":"https://images.unsplash.com/photo-1550547660-d9450f859349"},{"id":3,"name":"Pasta Alfredo","price":900,"category":"Italian","image":"https://images.unsplash.com/photo-1589308078059-be1415eab4c3"},{"id":4,"name":"Chocolate Shake","price":350,"category":"Beverages","image":"https://images.unsplash.com/photo-1572490122747-3968b75cc699"}]

App Screenshots:


Camera
Alina's Cafe
Fresh & Delicious
Rs. 550
Add to Cart
Pasta Alfredo
Rs. 900
Add to Cart
Chocolate Shake
Rs. 350
Add to Cart
Home
Explore
cart
manager


1:58
Camera
Place Order


Order Placed!
OK
Home
Explore
cart
manager


Camera
Alina's Cafe
Zinger Burger
Rs. 550
Add to Cart
Success
Order Sent!
OK
Rs. 900
Add to Cart
Home
Explore

Conclusion:

This project demonstrates full-stack development skills including frontend UI design, backend API development, and client-server integration. The application successfully implements mobile ordering functionality with modern UI design.